Topic: Partial fractions decompositions

Question: Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{5x + 3}{(x + 3)(x - 3)}$$

Answer choices:

A $$f(x) = -\frac{3}{x - 3} - \frac{2}{x + 3}$$

B $$f(x) = \frac{3}{x - 3} - \frac{2}{x + 3}$$

C $$f(x) = \frac{2}{x + 3} - \frac{3}{x - 3}$$

D $$f(x) = \frac{2}{x + 3} + \frac{3}{x - 3}$$

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Solution: D

These are distinct linear factors.

$$\frac{5x+3}{(x+3)(x-3)} = \frac{A}{x+3} + \frac{B}{x-3}$$

To solve for A, remove the $x+3$ factor and set $x=-3$ to find the value of the left side of the decomposition equation.

$$\frac{5x+3}{x-3} \to \frac{5(-3)+3}{-3-3} \to \frac{-12}{-6} \to 2$$

To solve for B, we'll remove the $x-3$ factor and set $x=3$.

$$\frac{5x+3}{x+3} \to \frac{5(3)+3}{3+3} \to \frac{18}{6} \to 3$$

Plugging $A=2$ and $B=3$ back into the partial fractions decomposition gives

$$f(x) = \frac{A}{x+3} + \frac{B}{x-3}$$

$$f(x) = \frac{2}{x+3} + \frac{3}{x-3}$$

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Topic: Partial fractions decompositions

Question: Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 - 8x^2 - 1}{(x+3)(x-2)(x^2+1)}$$

Answer choices:

A $$f(x) = \frac{2}{x+3} + \frac{1}{x-2} + \frac{1}{x^2+1}$$

B $$f(x) = \frac{2}{x+3} - \frac{1}{x-2} - \frac{1}{x^2+1}$$

C $$f(x) = \frac{2}{x+3} - \frac{1}{x-2} + \frac{1}{x^2+1}$$

D $$f(x) = \frac{2}{x+3} + \frac{1}{x-2} - \frac{1}{x^2+1}$$

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Solution: B

These are distinct linear and quadratic factors.

$$\frac{x^3-8x^2-1}{(x+3)(x-2)(x^2+1)}=\frac{A}{x+3}+\frac{B}{x-2}+\frac{Cx+D}{x^2+1}$$

To solve for A, remove the $x+3$ factor and set $x=-3$ to find the value of the left side of the decomposition equation.

$$\frac{x^3-8x^2-1}{(x-2)(x^2+1)}\to\frac{(-3)^3-8(-3)^2-1}{(-3-2)((-3)^2+1)}\to\frac{-100}{-50}\to 2$$

To solve for B, we'll remove the $x-2$ factor and set $x=2$.

$$\frac{x^3-8x^2-1}{(x+3)(x^2+1)}\to\frac{2^3-8(2)^2-1}{(2+3)(2^2+1)}\to\frac{-25}{25}\to -1$$

To solve for C, we'll multiply both sides of the equation by the denominator from the left side.

$$x^3-8x^2-1=A(x-2)(x^2+1)+B(x+3)(x^2+1)+(Cx+D)(x+3)(x-2)$$

$$x^3-8x^2-1=A(x^3+x-2x^2-2)+B(x^3+x+3x^2+3)+(Cx+D)(x^2+x-6)$$

$$x^3-8x^2-1=(A+B+C)x^3+(-2A+3B+C+D)x^2$$

$$+(A+B-6C+D)x+(-2A+3B-6D)$$

Equating coefficients gives us a system of equations.

$$A+B+C=1$$

$$-2A+3B+C+D=-8$$

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$$A + B - 6C + D = 0$$

$$-2A + 3B - 6D = -1$$

Solving $A + B + C = 1$ for A gives $A = 1 - B - C$, so the system simplifies to

$$5B + 3C + D = -6$$

$$7C - D = 1$$

$$5B + 2C - 6D = 1$$

Solving $7C - D = 1$ for D gives $D = 7C - 1$, so the system simplifies to

$$B + 2C = -1$$

$$B - 8C = -1$$

Subtracting the second equation from the first gives

$$B + 2C - (B - 8C) = -1 - (-1)$$

$$B + 2C - B + 8C = -1 + 1$$

$$10C = 0$$

$$C = 0$$

Then $D = -1$, $B = -1$, and $A = 2$. Plugging $A = 2$, $B = -1$, $C = 0$, and $D = -1$ back into the partial fractions decomposition gives

$$f(x) = \frac{A}{x+3} + \frac{B}{x-2} + \frac{Cx + D}{x^2 + 1}$$

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$$f(x) = \frac{2}{x+3} - \frac{1}{x-2} - \frac{1}{x^2+1}$$

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Topic: Partial fractions decompositions

Question: Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{4x^2 + 10x + 8}{(x+1)^3}$$

Answer choices:

A $$f(x) = \frac{4}{x+1} + \frac{2}{(x+1)^2} + \frac{2}{(x+1)^3}$$

B $$f(x) = \frac{4}{x+1} - \frac{2}{(x+1)^2} + \frac{2}{(x+1)^3}$$

C $$f(x) = \frac{4}{x+1} + \frac{2}{(x+1)^2} - \frac{2}{(x+1)^3}$$

D $$f(x) = \frac{4}{x+1} - \frac{2}{(x+1)^2} - \frac{2}{(x+1)^3}$$

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Solution: A

These are repeated linear factors.

$$\frac{4x^2+10x+8}{(x+1)^3}=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{(x+1)^3}$$

To solve for A, we'll multiply both sides of the equation by the denominator from the left side.

$$4x^2+10x+8=A(x+1)^2+B(x+1)+C$$

$$4x^2+10x+8=Ax^2+(2A+B)x+(A+B+C)$$

Equating coefficients gives us $A=4$ and then a system of equations.

$$2A+B=10$$

$$A+B+C=8$$

With $A=4$, we find $B=2$ and then $C=2$. Plugging $A=4$, $B=2$, and $C=2$ back into the partial fractions decomposition gives

$$f(x)=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{(x+1)^3}$$

$$f(x)=\frac{4}{x+1}+\frac{2}{(x+1)^2}+\frac{2}{(x+1)^3}$$

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